

Pedalboard Blueprint

Order, power and cabling without the hum

The pedal order most boards settle on and the reason for every position, the famous exceptions, how an effects loop changes things, and the power and noise checklists that fix most hum.

INSIDE

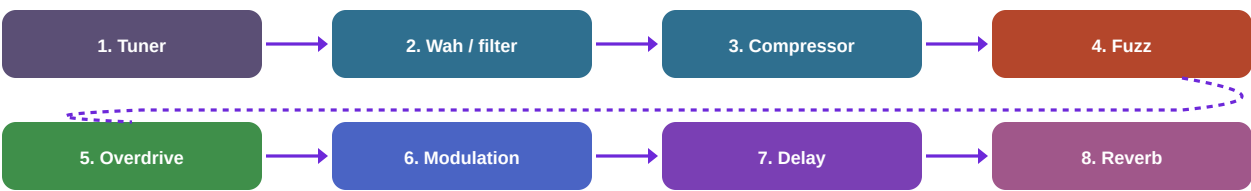
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Free from guitarhub.org. Share it, print it, pin it above the amp. The interactive versions of these ideas live at guitarhub.org/tone.

ORDER

The conventional order

Shape the raw signal, then distort it, then decorate the distorted sound with movement and space.



Guitar to amp, left to right, top row then bottom row.

Position	Why it goes there
Tuner	Sees the cleanest signal, and many can mute the whole board while you tune.
Wah and filters	Sweeping before gain makes the distortion itself sweep, which sounds vocal.
Compressor	Shapes your pick attack. After gain it mostly raises noise.
Fuzz	Many vintage fuzz circuits expect to see the guitar pickup directly and sound thin behind a buffer.
Overdrive, distortion	The core sound. Stack low gain into higher gain for focus.
Modulation	After gain, the movement stays clear instead of being squashed.
Delay, then reverb	Repeats that land in a room, rather than a room that echoes.

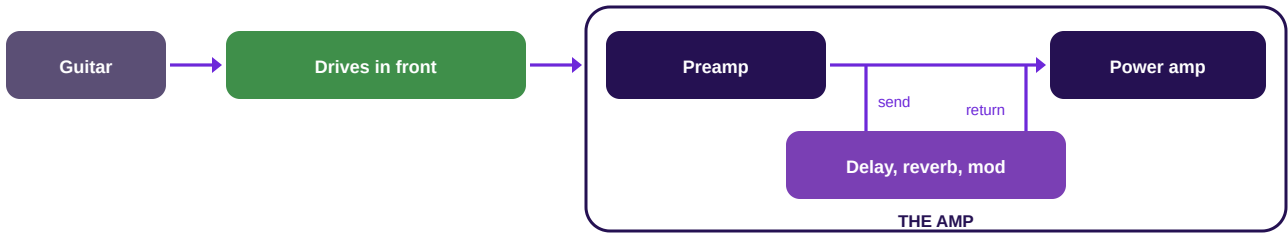
EXCEPTIONS

When to break the rules

Move	What you get
Reverb before drive	A distorted wash where the tail is as loud as the note. Shoegaze and noise rock.
Delay before drive	Repeats that grind into each other. Vintage tape-echo-into-amp sounds.
Wah after drive	A sharper, more synth-like filter sweep.
Phaser before drive	A subtler, throatier swirl. Common on classic hard rock.
Compressor after drive	Evens out level between clean and driven sounds; mostly used as a final leveller.
Swap two drives	Always try both orders. There is no correct answer, only the one you like.

The effects loop

Amps with preamp distortion often have a send and return between preamp and power amp. Time and modulation effects there sit after the amp's own distortion, which keeps them clean. Drives still go in front.



A four-cable setup: drives in front, time effects in the loop.

If your amp is clean and pedals make the distortion, you do not need the loop.

POWER

Power supply worksheet

Read every pedal's label. Most want 9 volts DC, center negative, but not all. The wrong voltage or polarity can damage a pedal.

Pedal	Volts	Polarity	Draw (mA)	Supply output	Output max (mA)
Total					

Isolated vs daisy chain. A daisy chain shares one ground and one supply, so digital pedals can leak whine into analog ones. Isolated outputs are separate circuits and solve most power noise.

Headroom. Give every pedal an output rated above its draw. Digital delays and reverbs usually draw far more than analog drives. Some pedals list a higher start-up draw; plan for it.

The noise hunt

Work from the amp outward and change one thing at a time, so you know which change fixed it.

- 1 Guitar straight into the amp. If it hums, rotate yourself and the guitar; single-coils pick up hum from lights, screens and transformers.
- 2 Add pedals back one at a time, all bypassed, then all on.
- 3 When the noise appears, move that pedal to its own isolated output.
- 4 Wiggle each patch cable while listening. Crackle means replace it.
- 5 Separate power and audio cables. Where they must cross, cross at right angles.
- 6 Check total cable length. A long true-bypass chain loses treble; one good buffer early in the chain restores it. Keep vintage fuzz before the buffer.
- 7 High-gain hiss is not a fault. A noise gate after the drives, or less gain, is the fix.

Board-day checklist

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| ☐ Spare patch cable | ☐ Tuner calibrated to A440 |
| ☐ Spare instrument cable | ☐ Velcro and zip ties |
| ☐ Spare power cable and adapter | ☐ Every pedal tested on and off |
| ☐ 9V battery for emergencies | ☐ Master volume levels matched |